Mōri: A Video Game Made in Unity

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ABSTRACT

Mōri is a first-person project developed in the Unity game engine. Using layered component-based architecture, the game's emphasises modularity, extensibility, and maintainability for the user and their choice of computer operating system. The system includes input handling, gameplay logic, AI behavior, UI management, and high-quality audio - all separated for organization and optimization with the layered component-based design. Taking heavy inspiration from the combat genre, *Mōri* has evolved from its early design to now a round-based combat game that mirrors the retro arcade feel of older titles like *DOOM*. Major functions such as a minimalistic user interface, complex enemy AI, inventory, currency systems, as well as a map and round overlay help immerse the player.

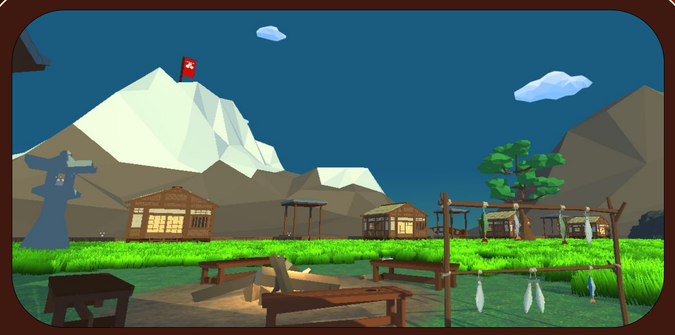

Figure 1: One of the maps created for Mōri; the map design is inspired by medieval Japanese media.

CONCLUSION

Our team had the opportunity to design and develop Mōri from the ground up, gaining hands-on experience with Unity's component-based architecture, AI behavior systems, and game UI design. Through iterative development and peer feedback, we produced a functional, polished game that reflects our growth as software engineers. Mōri stands as a testament to our team's dedication to creating an immersive, high-quality experience.

DESCRIPTION

THE YEAR IS 1578, AND JAPAN IS AT WAR…

As Mōri Katsuko, your objective is to storm the Oda clan capitol and defeat their daimyo. You must hurry, for the Oda clan army is returning, and their numbers shall surely envelop you.

On this quest, you can interact with shopkeepers who will provide you with the necessary weaponry and utilities to overcome the enemies you encounter on your journey. But, money can only keep you alive for so long, so look for the various power-ups that drop randomly as you slay your enemies. These power-ups will enhance your abilities by adding elements like damage multipliers, attack speed boosts, and special attacks.

The knowledge and skills you gain during your travels through feudal Japan will reveal their usefulness as you make it to the castle and challenge the Oda daimyo to a duel. Your face-off against them shall be the ultimate test of your skills and strength. Will you succeed and bring honor to your clan, or will your clan lie with you in the grave in shame?

CHALLENGES

One massive challenge we faced during development was the incorporation of smart, fast-acting enemies to challenge the player. The combat system for this was constantly revised, often improving (or limiting) the defensive and offensive capabilities of the enemy troops. This issue bled into our other level components, such as the player's health and movement systems, to even the interactive banner feature that you must use to progress through the rounds. Many times, the enemies felt easy to defeat, while other times felt as if they could instantly swarm and kill the player. We wanted a steady progression of the difficulty between each round - random enemy capabilities was certainly going against that. Fortunately, through our animation and programming re-designs, we managed to find a solid balance that can dynamically change based off the current round being played.

Figure 2: A group of enemies can easily swarm you and keep you staggered with constant attacks.

Figure 3: The game will lock the player into the round until they defeat all the enemies.

This project was developed during the Spring 2026 semester as part of the *Computer Science 426 - Senior Projects* course in Computer Science.

FUTURE WORK

Mōri is a project that continues to evolve, as there is ample opportunity to build on the narrative and gameplay.

1. Implementation of a single player campaign, wherein you must partake in a castle siege. This campaign will have multiple distinct maps, each with their own enemy types and boss battles.
2. Add more depth to the combat system by giving players the ability use more than one weapon, throwable objects and a wider array of special abilities.
3. Update the shop and exploration systems of the game, which can provide the player better odds of survival through the act of finding, or purchasing, consumables.

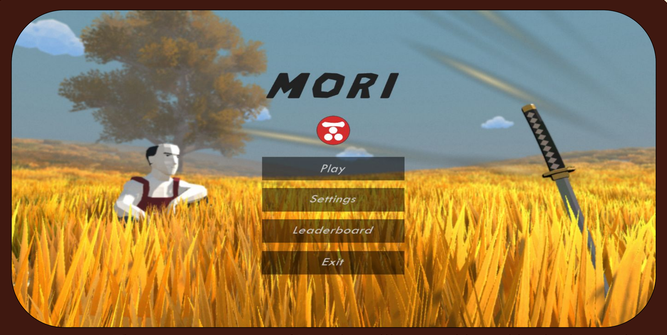

Figure 4: The main menu screen of Mōri that will come with soothing background music and some 3D models, symbolizing the somber reality of the player's daunting objective.

ACKNOWLEDGEMENTS

We would like to thank our instructors, advisors, and our classmates for devoting their time to our project. We appreciate them for providing vital feedback on our gameplay so that we, in turn, may produce a well-designed, high-quality game that is unique amongst the other world-renown combat games *Mōri* takes inspiration from.